

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Agentic AI systems with execution-grounded evaluation – Multimodal model routing & serving – Production ML at enterprise scale

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a closed-loop agentic pipeline for parametric CAD code generation: a frozen LLM generates CadQuery programs and execution feedback (compile validity, geometric correctness via Chamfer/Hausdorff on STL/B-Rep) drives a LangGraph-orchestrated repair loop – inference-time search with verifiable geometric reward; 16-module system maintained as a 202-test automated evaluation harness with VLM-based visual judgment.
- Designed scalable data curation and processing for the pipeline’s retrieval-augmented prompting: structured evidence collection, prompt-evaluation components, and failure classification feeding the automated quality-control gates of a 202-test evaluation harness.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ARTEMIS** (Aug 2025–Present): Developing cost-aware routing for Vision-Language Models (VLMs): trained neural multi-task router selects the optimal VLM per request across VQA, OCR, captioning, and reasoning – SLA-aware load balancing serves 6+ heterogeneous backends (Gemma 3, Qwen-VL, DeepSeek OCR, Llama-4) via vLLM with 5 dynamic routing modes.
- **CERBERUS** (Aug 2025–Present): Encoder-frozen cross-modal alignment research: trained alignment stack with PyTorch DDP + mixed precision on H100/A100 clusters; Matryoshka Representation Learning compresses embeddings 4096→128 dims at 96% retention – R@5 78% on PixMo vision-text retrieval; ablated Perceiver Resampler integration and diagnosed retrieval collapse to initialization/gradient issues.
- **ATHENA** (Jan–May 2025): Built a 5-agent supervisor-worker multi-agent system via LangGraph Command routing with reflection-based critique; designed cross-modal evaluation framework (BLEU-4, METEOR, BERTScore, CLIPScore) over 100 reference videos – +0.18 BLEU-4 absolute over single-agent baseline; ablation cut fact coverage 27% without the critique agent.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led development of an agentic RAG diagnostics system from hackathon prototype to production deployment: LLM plans multi-step tool calls over network telemetry and verifies hypotheses via structured function calling – autonomous root-cause analysis reducing mean resolution time 70%.
- Scaled OpenSearch telemetry ingestion pipelines from 50 to 2,000 events/sec (40x) via complete re-architecture (async I/O + Golang); deployed edge ML inference via ONNX Runtime (40% latency reduction); built SLA forecasting for 60+ production classifiers across 4 categories with automated evaluation and retraining.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection; reduced manual troubleshooting 75%. Security-telemetry ML deployed in a production network-security environment (patented anomaly detection).

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Distributional thresholding for connectivity anomaly detection.

IEEE ICAIA 2026: ML + Memory Forensics for IoT Cybersecurity (co-author).

TECHNICAL SKILLS

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, Transformers, NLP, Computer Vision, Model Training & Evaluation, Scikit-Learn, NumPy, Pandas, Experimentation & Ablation Design

Agentic AI: LangChain, LangGraph, Multi-Agent Frameworks, Tool/Function Calling, Agentic RAG, Planning & Reflection, Long-Horizon Task Decomposition, Structured Generation (Pydantic)

Inference & Deployment: vLLM, Inference Optimization (ONNX Runtime), Latency/Throughput Engineering, Model Serving, Distributed Training (PyTorch DDP, Mixed Precision), Docker, Linux, Azure

Data Pipelines: Python, Go, SQL/PostgreSQL, OpenSearch/Elasticsearch, Redis, Data Curation & Quality Control, ETL, FAISS

Security ML: Adversarial Robustness (PGD, Poisoning, Membership Inference – coursework), Anomaly Detection (patented), Memory Forensics (published)

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

EDUCATION

Georgia Institute of Technology <i>M.S. in Computer Science (Machine Learning Specialization), GPA: 4.0/4.0</i> – Coursework: Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI	Atlanta, GA <i>Aug 2024 – Dec 2026</i>
Maharaja Surajmal Institute of Technology <i>B.Tech. in Information Technology, CGPA: 8.8/10.0</i>	New Delhi, India <i>Aug 2016 – Aug 2020</i>